

Computer and Network Security

Rapid-Revision One-Liners

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COMPLETE PREPARATION SERIES

100% SYLLABUS COVERAGE
CONCEPT-FOCUSED LEARNING
MCQ-ORIENTED APPROACH
QUICK REVISION READY

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2026 EDITION
VERSION 1.0

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Computer and Network Security — One-Liners

Security Fundamentals

1. The **CIA triad** has **3** security goals: **Confidentiality**, **Integrity** and **Availability**.
2. **Confidentiality** is achieved by **encryption**, while **integrity** is achieved by **hashing/MAC** and **availability** by **redundancy/backups**.
3. There are **2** goals beyond the core triad that are commonly added: **authentication** and **non-repudiation**.
4. **Non-repudiation** is provided by exactly **1** mechanism: the **digital signature**.
5. Risk is modelled by **3** factors in the relation **Risk = Threat × Vulnerability × Impact**.
6. A **vulnerability** is a **weakness**, whereas a **threat** is a **potential cause** of an unwanted incident.
7. Security attacks split into **2** classes: **passive attacks** and **active attacks**.
8. There are **2** types of passive attack: **release of message contents** and **traffic analysis**.
9. There are **4** types of active attack: **masquerade**, **replay**, **modification** and **denial of service (DoS)**.
10. **Passive** attacks are hard to **detect** but easy to **prevent**, so the emphasis is on **prevention**.
11. **Active** attacks are hard to **prevent** but easy to **detect**, so the emphasis is on **detection and recovery**.
12. **Interception** violates **confidentiality**, and it is classified as a **passive** attack.
13. **Interruption** violates **availability**, and it is classified as an **active** attack.
14. **Modification** violates **integrity**, while **fabrication** violates **authenticity**.
15. The **X.800** standard defines **5** security service categories: **authentication**, **access control**, **data confidentiality**, **data integrity** and **non-repudiation**.
16. **X.800** lists **8** specific security mechanisms including **encipherment**, **digital signature**, **access control**, **data integrity**, **authentication exchange**, **traffic padding**, **routing control** and **notarization**.
17. The only X.800 mechanism that directly provides confidentiality is **1**: **encipherment** (encryption).
18. **Masquerade** is an active attack in which **1** entity pretends to be a **different** entity.

Cryptography Basics

19. **Cryptology** is the umbrella field that combines **2** disciplines: **cryptography** (making ciphers) and **cryptanalysis** (breaking ciphers).
20. **Encryption** converts **plaintext** into **ciphertext**, while **decryption** reverses it back to **plaintext**.
21. There are **4** classic cryptanalytic attack models: **ciphertext-only**, **known-plaintext**, **chosen-plaintext** and **chosen-ciphertext**.
22. The hardest cryptanalytic attack for the adversary is **1**: the **ciphertext-only** attack.
23. A brute-force attack on an **n-bit** key needs on average **$2^{(n-1)}$** trials, i.e. half of **2^n** .
24. **Kerckhoffs's principle** states that security must rest on **1** secret only: the **key**, never the algorithm.
25. Only **1** cipher is provably **unconditionally secure**: the **One-Time Pad**, proved by **Claude Shannon**.

Classical Encryption Techniques

26. The **Caesar cipher** shifts letters by a fixed **3** (classic) and has only **25** usable keys via **$C = (P+k) \bmod 26$** .
27. A **monoalphabetic** cipher has **$26!$** (**$\sim 4 \times 10^{26}$**) keys yet is broken by **1** technique: **frequency analysis**.
28. The **Playfair cipher** encrypts **2** letters at a time (digraphs) using a **5×5** matrix in which **I and J** share one cell.
29. The Playfair cipher was invented by **Charles Wheatstone** but named after **Lord Playfair**.
30. The **Hill cipher** is polygraphic and uses **matrix multiplication mod 26**, requiring the key matrix to be **invertible**.
31. The **Vigenère cipher** is **polyalphabetic**, using a repeating **keyword**, and is broken by the **Kasiski examination**.
32. The **One-Time Pad** needs a key that is **truly random**, **as long as the message**, and used exactly **1** time.
33. Substitution ciphers **replace** symbols, while transposition ciphers **rearrange** their positions.
34. There are **2** classic transposition ciphers in the syllabus: the **Rail Fence** and the **Columnar** transposition.
35. The **Rail Fence** cipher writes plaintext in a **zigzag** across **N rails** then reads it **row by row**.
36. **Steganography** hides the **existence** of a message, whereas cryptography hides only its **meaning**.
37. A common digital steganography method is **1**: **LSB (Least Significant Bit)** embedding in images.

Symmetric Key Cryptography

38. **Symmetric** cryptography uses exactly **1** shared key for both encryption and decryption.
39. For **n** users, symmetric systems need $n(n-1)/2$ keys, which is the **key distribution** problem.
40. **Block ciphers** process fixed **blocks** (64/128 bits), while **stream ciphers** process **1 bit/byte** at a time.
41. Stream ciphers encrypt by **XOR-ing** plaintext with a **keystream**, making them generally **faster** than block ciphers.
42. A **Feistel** cipher splits each block into **2** halves (**Left** and **Right**) and uses the same structure to **decrypt**.
43. The Feistel round equation uses **1** function **F**: $R_i = L_{(i-1)} \text{ XOR } F(R_{(i-1)}, K_i)$ with $L_i = R_{(i-1)}$.
44. **DES** has a block size of **64 bits** and a key size of **56 effective bits** (64 with parity).
45. **DES** runs **16** rounds, each using a **48-bit** subkey, and uses **8** S-boxes that map **6 bits to 4 bits**.
46. **DES** became insecure because its 2^{56} key space was brute-forced in **1998**.
47. **Triple DES (3DES)** applies DES **3** times in **E-D-E** order as $C = E(K_3, D(K_2, E(K_1, P)))$.
48. **2-key 3DES** gives a **112-bit** key, while **3-key 3DES** gives a **168-bit** key.
49. **Double DES** is avoided because **1** attack halves its strength: the **meet-in-the-middle** attack.
50. **AES** has a fixed block size of **128 bits** and supports **3** key sizes: **128, 192** and **256** bits.
51. AES key sizes **128/192/256** map to **10/12/14** rounds respectively.
52. **AES** is based on the **Rijndael** cipher and is a **Substitution-Permutation Network (SPN)**, not a Feistel cipher.
53. Each AES round has **4** steps: **SubBytes**, **ShiftRows**, **MixColumns** and **AddRoundKey**.
54. The **final AES round** omits exactly **1** step: **MixColumns**.
55. AES operates on a **4x4** state matrix of **16** bytes (= 128 bits).
56. There are **5** standard block-cipher modes: **ECB, CBC, CFB, OFB** and **CTR**.
57. The least secure mode is **1**: **ECB**, because identical plaintext blocks yield identical ciphertext blocks.
58. **CBC** mode XORs each block with the **previous ciphertext** and requires an **IV (Initialization Vector)**.
59. **3** modes turn a block cipher into a stream cipher: **CFB, OFB** and **CTR**.
60. **CTR** mode encrypts an incrementing **counter** and is fully **parallelizable** for both encryption and decryption.
61. **RC4** is a **stream** cipher by **Ron Rivest** that uses a **256-byte** state array with a **KSA** and **PRGA**.
62. **RC4** was used in **2** notable protocols now deprecated: **WEP** and early **SSL/TLS**.

Public Key (Asymmetric) Cryptography

63. **Asymmetric** cryptography uses **2** mathematically linked keys: a **public key** and a **private key**.
64. For **n** users, asymmetric systems need only **2n** keys, versus $n(n-1)/2$ for symmetric.
65. For **confidentiality** you encrypt with the **receiver's public** key and decrypt with their **private** key.
66. For **authentication/signing** you use the **sender's private** key and verify with the **sender's public** key.
67. **RSA** was published in **1977** by **3** people: **Rivest, Shamir and Adleman**.
68. **RSA** security rests on **1** hard problem: **factoring** the large modulus $n = p \times q$.
69. In RSA, $\phi(n) = (p-1)(q-1)$ and the public exponent **e** must satisfy $\text{gcd}(e, \phi(n)) = 1$.
70. In RSA the private key is computed as **1** value: $d = e^{-1} \text{ mod } \phi(n)$.
71. RSA encryption is $C = M^e \text{ mod } n$ and decryption is $M = C^d \text{ mod } n$.
72. The RSA public key is the pair **(e, n)** while the private key is the pair **(d, n)**.
73. **Diffie-Hellman** (1976) performs **1** function only — **key exchange** — and does **not** encrypt or sign.
74. Diffie-Hellman security rests on **1** hard problem: the **discrete logarithm** problem.
75. In Diffie-Hellman the shared secret is $K = a^{(xy)} \text{ mod } q$, computed from public values a^x and a^y .
76. **Diffie-Hellman** is vulnerable to **1** major attack because it lacks authentication: the **Man-in-the-Middle** attack.
77. **ElGamal** is based on the **discrete logarithm** problem and produces ciphertext that is **2x** the plaintext size.
78. **ECC** relies on **1** problem — the **Elliptic Curve Discrete Logarithm Problem (ECDLP)** — and uses much smaller keys.
79. A **256-bit ECC** key gives roughly the same security as a **3072-bit RSA** key.
80. Asymmetric crypto is roughly **1000x slower** than symmetric, so real systems use **hybrid** encryption.
81. **3** common asymmetric algorithms are **RSA, Diffie-Hellman** and **ECC**, versus symmetric **DES/AES/RC4**.

PKI and Digital Certificates

- 82. A **digital certificate** binds a **public key** to an **identity** and is digitally signed by **1** entity: the **CA**.
- 83. The certificate standard is **X.509**, whose fields include **version**, **serial number**, **issuer**, **validity** and **subject public key**.
- 84. In PKI the **CA** issues certificates, while the **RA** only **verifies identity** and does not issue.
- 85. Certificate revocation is checked by **2** mechanisms: the **CRL (Certificate Revocation List)** and **OCSP**.
- 86. **OCSP** stands for the **Online Certificate Status Protocol** and checks **1** certificate's status in real time.

Cryptographic Hash Functions

- 87. A cryptographic hash maps arbitrary input to a **fixed-length** digest and is a **one-way** function.
- 88. Hash functions require **3** security properties: **preimage**, **second-preimage** and **collision** resistance.
- 89. **Preimage resistance** means it is infeasible to find **1** input **m** for a given hash **h = H(m)**.
- 90. The **avalanche effect** means a **1-bit** input change flips about **half** the output bits.
- 91. A **birthday attack** breaks an **n-bit** hash in about **2^(n/2)** work by targeting **collision** resistance.
- 92. **MD5** produces a **128-bit** digest with a **512-bit** block and is now **collision-broken**.
- 93. **SHA-1** produces a **160-bit** digest and was broken in **2017** by Google's **SHAttered** attack.
- 94. **SHA-256** produces a **256-bit** digest with a **512-bit** block, while **SHA-512** produces **512 bits** with a **1024-bit** block.
- 95. The **SHA-2** family contains **4** main variants: **SHA-224**, **SHA-256**, **SHA-384** and **SHA-512**.
- 96. **SHA-3** uses **1** distinct construction — the **sponge (Keccak)** — and was standardised in **2015**.
- 97. Hash functions provide **integrity** but **not** confidentiality, because they cannot be reversed.
- 98. A **MAC** provides **2** services — **authentication and integrity** — but **not** non-repudiation, since the key is shared.
- 99. **HMAC** is defined in **RFC 2104** as **H((K⊕opad) || H((K⊕ipad) || M))**.
- 100. In HMAC the **2** pad constants are **ipad = 0x36** and **opad = 0x5C**, each repeated.
- 101. Secure password storage uses **salted hashes** via **4** common KDFs: **bcrypt**, **PBKDF2**, **scrypt** and **Argon2**.

Digital Signatures

- 102. A **digital signature** provides **3** services: **authentication**, **integrity** and **non-repudiation**.
- 103. You **sign** a message with the **private** key and **verify** the signature with the **public** key.
- 104. For efficiency, a digital signature signs only **1** thing — the message **hash** — not the whole message.
- 105. **DSA** is part of the **DSS (FIPS 186)** standard and is based on the **discrete logarithm** problem.
- 106. **DSA** performs **1** operation only — **signing** — and cannot perform encryption, unlike **RSA**.
- 107. Compared to RSA, **DSA** has **faster signing** but **slower verification**.
- 108. The elliptic-curve form of DSA is **1** algorithm: **ECDSA**, which uses small keys.
- 109. A digital **signature** authenticates a **message**, whereas a digital **certificate** authenticates a **public key's owner**.
- 110. **Non-repudiation** is provided by **digital signatures** but **not** by **MACs**.

Authentication

- 111. There are **3** authentication factors: something you **KNOW**, something you **HAVE** and something you **ARE**.
- 112. A **password/PIN** is the **knowledge** factor, a **token/smart-card** is the **possession** factor, and **biometrics** is the **inherence** factor.
- 113. **2FA** requires **2** factors from **different** categories, so password + PIN does **not** qualify (both are "know").
- 114. Biometric accuracy uses **2** error rates — **FAR (False Acceptance Rate)** and **FRR (False Rejection Rate)** — equal at the **CER/EER**.
- 115. There are **2** OTP standards: **HOTP** (counter-based, RFC 4226) and **TOTP** (time-based, RFC 6238).
- 116. A **rainbow table** attack uses precomputed hash tables and is defeated by **1** countermeasure: the **salt**.
- 117. A **salt** is a random per-user value, while a **pepper** is a **secret** salt stored separately.
- 118. A **dictionary attack** tries **common words**, whereas **brute force** tries **every** possible combination.
- 119. **Kerberos** uses **symmetric** key cryptography and runs on default port **88**.

- 120. The Kerberos **KDC** has **2** components: the **AS (Authentication Server)** and the **TGS (Ticket-Granting Server)**.
- 121. Kerberos first issues **1** ticket — the **TGT (Ticket-Granting Ticket)** — before granting service tickets.
- 122. Kerberos defeats **replay** attacks using **timestamps**, which is why it needs **clock synchronisation**.
- 123. Kerberos was developed at **MIT** under **Project Athena**, and the current version is **v5 (RFC 4120)**.
- 124. **Challenge-response** authentication sends a random **nonce** so the secret is **never** transmitted, defeating replay.

Network Security Protocols

- 125. **SSL** evolved into **TLS**, where **SSL 3.0** became **TLS 1.0**, and the current versions are **TLS 1.2** and **TLS 1.3**.
- 126. **HTTPS** is **HTTP over TLS** and runs on port **443**.
- 127. TLS uses **asymmetric** crypto for the **handshake/key exchange** and **symmetric** crypto for **bulk data** (hybrid).
- 128. The TLS handshake establishes **1** shared secret — the **master secret** — used to derive session keys.
- 129. TLS has **4** sub-protocols: **Handshake**, **Record**, **Change Cipher Spec** and **Alert**.
- 130. **IPSec** operates at the **network (IP) layer** and uses **2** protocols: **AH** and **ESP**.
- 131. **AH** is IP protocol **51** and provides **integrity + authentication** but **no** confidentiality (no encryption).
- 132. **ESP** is IP protocol **50** and provides **confidentiality + integrity + authentication** (it encrypts).
- 133. IPSec has **2** modes: **transport mode** (payload only) and **tunnel mode** (whole packet, used for **VPNs**).
- 134. IPSec key management uses **IKE (Internet Key Exchange)** on **UDP port 500**.
- 135. An IPSec **Security Association (SA)** is identified by **1** value: the **SPI (Security Parameter Index)**.
- 136. **SSH** secures remote login on port **22** and replaces insecure **Telnet** on port **23**.
- 137. The current SSH version is **SSH-2**, and it supports both **password** and **public-key** authentication.
- 138. **PGP** was created by **Phil Zimmermann** in **1991** and uses a decentralised **web of trust** (no CA).
- 139. **PGP** is **hybrid**, using a **session key** (IDEA/3DES/AES) plus **RSA/DH** key wrapping and **ZIP** compression.
- 140. **S/MIME** secures email using **X.509** certificates and a **hierarchical CA** model, unlike PGP's web of trust.
- 141. A **VPN** creates an encrypted **tunnel** over a public network using protocols like **IPSec**, **L2TP**, **PPTP** and **OpenVPN**.
- 142. **PPTP** runs on port **1723** but is considered **weak/insecure** today.
- 143. There are **2** VPN topologies: **site-to-site** (network-to-network) and **remote-access** (user-to-network).

Firewalls and Network Defense

- 144. A **firewall** filters traffic using a **rule set/ACL** to enforce a perimeter security policy.
- 145. A **packet-filtering** firewall works at **L3/L4** and is **stateless**, judging each packet alone.
- 146. A **stateful inspection** firewall tracks the **connection state table**, unlike a stateless packet filter.
- 147. An **application gateway/proxy** firewall works at **Layer 7** and inspects application **content**.
- 148. A **circuit-level gateway** works at **Layer 5** and validates the **TCP session/handshake**.
- 149. **IDS** is **passive** and only **detects + alerts**, while **IPS** is **inline/active** and **blocks** traffic.
- 150. IDS/IPS use **2** detection methods: **signature-based** (misuse) and **anomaly-based** (behaviour).
- 151. **Signature-based** detection cannot catch **zero-day** attacks, while **anomaly-based** detection causes more **false positives**.
- 152. A **false positive** flags benign traffic as malicious, while a **false negative** misses a **real** attack.
- 153. IDS deploys in **2** forms: **NIDS** (network-based) and **HIDS** (host-based).
- 154. A **DMZ** is a perimeter subnet hosting public servers and is typically built with **2** firewalls (screened subnet).
- 155. Antivirus detects malware by **3** methods: **signatures**, **heuristics** and **behaviour monitoring**.

Malware

- 156. A **virus** needs a **host file** and **human action** to spread, unlike a worm.
- 157. A **worm** is **standalone** and **self-replicating**, spreading automatically over the **network**.
- 158. A **Trojan** is **disguised** as legitimate software and does **not** self-replicate.
- 159. **Ransomware encrypts** the victim's data and demands a **ransom** for the decryption key.
- 160. **Spyware** secretly **gathers information**, while **adware** displays unwanted **advertisements**.

- 161. A **rootkit hides** its presence and aims to gain **admin/root** privileges.
- 162. A **botnet** is a network of compromised **bots** controlled remotely, commonly used for **DDoS**.
- 163. A **logic bomb** triggers on **1** specific condition (date/event), while a **backdoor** provides hidden **access**.
- 164. A **keylogger** records **keystrokes**, capturing passwords and sensitive input.

Attacks (Network & Application)

- 165. A **Man-in-the-Middle (MITM)** attack secretly **relays/alters** traffic between **2** parties, defeated by **TLS/mutual authentication**.
- 166. A **replay attack** retransmits **captured** valid data and is defeated by **nonces, timestamps** and **sequence numbers**.
- 167. A **DoS** attack floods a target from **1** source, while a **DDoS** uses **many** sources (a **botnet**).
- 168. **3** classic DoS techniques are the **SYN flood**, the **Smurf** (ICMP) attack and the **Ping of Death**.
- 169. A **SYN flood** abuses the **TCP 3-way handshake** by sending many **SYN** packets without completing them.
- 170. **Phishing** has targeted variants: **spear phishing** (specific person), **whaling** (executives), **vishing** (voice) and **smishing** (SMS).
- 171. **SQL Injection** inserts malicious **SQL** via input and is defeated by **parameterized queries** and input validation.
- 172. A classic SQLi payload is **1** string — ' OR '1'='1 — which makes the WHERE clause always **true**.
- 173. **XSS** injects **client-side scripts** into pages and has **3** types: **stored, reflected** and **DOM-based**.
- 174. **XSS** is defended by **output encoding**, input validation and a **Content Security Policy (CSP)**.
- 175. **CSRF** forces an authenticated user's browser to send an unwanted request, defeated by **anti-CSRF tokens** and **SameSite** cookies.
- 176. **SQLi** targets the **database (server-side)**, while **XSS** targets other users' **browsers (client-side)**.
- 177. **Spoofing** has **4** common forms: **IP, ARP, DNS** and **MAC** spoofing.
- 178. **ARP spoofing** poisons the **ARP cache** to perform **MITM** on a LAN.
- 179. **DNS spoofing/cache poisoning** redirects a victim to a **fake** website by corrupting DNS records.
- 180. **Sniffing** is the **passive** capture of traffic using tools such as **Wireshark** and **tcpdump**.
- 181. Sniffing is most dangerous on **unencrypted** protocols and on shared/**hub**-based networks.
- 182. **Social engineering** exploits **human** trust rather than technical flaws, using tactics like **pretexting, baiting** and **tailgating**.
- 183. **Credential stuffing** reuses **leaked** username/password pairs across multiple sites.
- 184. **IP spoofing** forges the **source IP address** to hide the attacker or bypass IP-based filters.
- 185. A **DDoS amplification** attack (DNS/NTP) uses small requests to trigger **large** responses toward the victim.